Math 55 worksheet Wednesday 02/04/2026

1. Russell's Paradox considers the set of all sets that don't contain themselves (i.e. $\{x : x \notin x\}$), and gets into a contradiction.

   Use Russell's Paradox to show that every set has a subset that is not an element of it.

2. Recall that a function $f : A \to B$ is *onto* iff every $b \in B$ gets "hit" by some $a \in A$, i.e. $f(a) = b$. Show that there is no onto function from a set X onto $\mathcal{P}(X)$.

3. Recall that a function $f : A \to B$ is *one-to-one* iff whenever $f(x) = f(y)$, we get $x = y$. Prove that there is no one-to-one function from the powerset of a set X into X.

4. Conclude that for any set X, we have $|X| < |\mathcal{P}(X)|$.